

cocotb BinaryValue

Plain Python integers do not carry bit width

Width, value, and bit string

- N bits sets how many bits are kept, wider input is truncated to the low bits
- Starter example
- Decimal two fifty-five and hex F F mean the same eight-bit pattern
- `Dut.data.value` equals `BinaryValue` with your literal and n bits equals width
- To read back, convert the signal value to int when you need a Python number

Browser lab

Digital Design and Verification Platform

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INTERACTIVE TOOL

cocotb BinaryValue

Width + integer/hex value → bit string and **BinaryValue** poke sketch. Starter: width 8, value 0xA5 → 10100101.

Starter example: width 8, value 0xA5 → bits 10100101 (MSB left).

Load starter example

Challenges 0 / 22 cleared

Quiz: width: n_bits on BinaryValue...

☐ limits the value to that many low bits

☐ sets cocotb log level

☐ replaces the clock period

☐ disables assertions

Show hint

Check

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Idle

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Core ideas

width

n_bits masks to that many bits.

value

Int or hex — truncated to width.

MSB

Leftmost char is bit (width-1).

poke

Assign .value to drive the signal.

Lab

Browser lab starter

learn_cocotb — cocotb binary value

Real cocotb practice

- In the real cocotb track, work the bit math on paper before you drive a live signal
- Write BinaryValue for hex A five with n bits eight and draw the MSB-left string bit by bit
- Try width four with hex F and confirm all ones
- Optional: open the lab's source sketch and read how poke and int peek are commented
- This module is width-aware drive literacy, not a full scoreboard yet

Pitfalls to watch

- Do not assume Python's bin of an integer matches cocotb's MSB-left string without applying
- A common miss is reading bits MSB-right, cocotb prints index width minus one on the left
- Another trap is poking a wider value than the port
- And remember

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, load the starter and predict the bit string before you set value
- On paper, convert hex A five to eight bits MSB left without looking
- When you are ready